

Diagnosis information PDK control unit

Shiftlock solenoid - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Selector lever control unit faulty

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Shiftlock function can no longer be carried out
- ♦ The selector lever is locked in position P and can only be unlocked mechanically (emergency release)
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated

P0931

Diagnosis information PDK control unit

Shiftlock solenoid - short circuit to ground, to B+ or open circuit

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Selector lever control unit faulty

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Shiftlock function can no longer be carried out
- ♦ The selector lever is locked in position P and can only be unlocked mechanically (emergency release)
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated

Diagnostic information - PDK control unit

Electric pressure adjuster 1 (clutch 1)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary current of low-side pressure adjuster 1 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 1 is switched off
- ♦ Low-side electric pressure adjuster 1 is switched off
- ♦ Only 2nd, 4th and 6th gears are possible
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ♦ If short circuit to B+, then all hydraulic control elements are switched off

► Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 1 (clutch 1)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary current of low-side pressure adjuster 1 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 1 is switched off
- ♦ Low-side electric pressure adjuster 1 is switched off
- ♦ Only 2nd, 4th and 6th gears are possible
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ♦ If short circuit to B+, then all hydraulic control elements are switched off

► Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 1 (clutch 1)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary current of low-side pressure adjuster 1 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 1 is switched off
- ♦ Low-side electric pressure adjuster 1 is switched off
- ♦ Only 2nd, 4th and 6th gears are possible
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ♦ If short circuit to B+, then all hydraulic control elements are switched off

► Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 2 (clutch 2)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 2 not energized
- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 2 (clutch 2)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 2 not energized
- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 2 (clutch 2)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 2 not energized
- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 3 (system pressure)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 3 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 3 not energized
- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 3 (system pressure)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 3 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 3 not energized
- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

Diagnostic information - PDK control unit

Electric pressure adjuster 3 (system pressure)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 3 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 3 not energized
- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

P0973

Diagnostic information - PDK control unit

Solenoid valve 1 (selection of gear train half)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Solenoid valve 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P0974

Diagnostic information - PDK control unit

Solenoid valve 1 (selection of gear train half)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Solenoid valve 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P0976

Diagnostic information - PDK control unit

Solenoid valve 2 (shift cylinder selection)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Solenoid valve 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P0977

Diagnostic information - PDK control unit

Solenoid valve 2 (shift cylinder selection)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Solenoid valve 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P1706

Diagnosis information PDK control unit

5-V sensor supply voltage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

- Sensor supply voltage > 5.5 V
- Sensor supply voltage < 4.5 V

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Low-side electrical pressure adjuster 5 is switched off
- ♦ Low-side electrical pressure adjuster 6 is switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Stop message from CAN is contradicted
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ All adaptations are deactivated
- ♦ Electrical pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ Pressure-sensor plausibility check is switched off
- ♦ Pressure control is switched off

Control unit setpoints

- ♦ Supply voltage for pressure sensor and distance sensor 5 V

P1707

Diagnosis information PDK control unit

8.5-V sensor supply voltage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

- Sensor supply voltage > 9.5 V
- Sensor supply voltage < 7 V

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Input-shaft speed sensors 1 and 2 assume a substitute value

Control unit setpoints

- ♦ Sensor supply voltage 8.5 V

P170A

Diagnosis information PDK control unit

5-V sensor supply voltage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

- Sensor supply voltage > 5.5 V
- Sensor supply voltage < 4.5 V

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Low-side electrical pressure adjuster 5 is switched off
- ♦ Low-side electrical pressure adjuster 6 is switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Stop message from CAN is contradicted
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ All adaptations are deactivated
- ♦ Electrical pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ Pressure-sensor plausibility check is switched off
- ♦ Pressure control is switched off

Control unit setpoints

- ♦ Supply voltage for pressure sensor and distance sensor 5 V

P170B

Diagnosis information PDK control unit

8.5-V sensor supply voltage

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

- Sensor supply voltage > 9.5 V
- Sensor supply voltage < 7 V

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Input-shaft speed sensors 1 and 2 assume a substitute value

Control unit setpoints

- ♦ Sensor supply voltage 8.5 V

P1715

Diagnosis information PDK control unit

PDK control unit - internal fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ PDK control unit faulty

Fault setting condition(s)

- Internal diagnostic routine returns fault

Fault effects

- ♦ Transmission fault display in the instrument cluster control unit
- ♦ Emergency stop function

P172B

Diagnosis information PDK control unit

Transmission temperature sensor - gradient error

Information

For this purpose, observe the Technical Information for model years 2013 and 2014: **3734 - PDK transmission: fault codes P172D, P17F1, P17F2, P18CA, P18CB (65/13)**

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

- If the gradient exceeds a threshold, the gradient error counter is incremented and the fault is entered.

Fault effects

- ♦ Electrical substitute value is assumed
- ♦ Substitute value is assumed
- ♦ All adaptations are deactivated
- ♦ No start/stop

P172D

Diagnosis information PDK control unit

Transmission temperature sensor - gradient error

Information

For this purpose, observe the Technical Information for model years 2013 and 2014: **3734 - PDK transmission: fault codes P172D, P17F1, P17F2, P18CA, P18CB (65/13)**

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines

Fault setting condition(s)

- If the gradient exceeds a threshold, the gradient error counter is incremented and the fault is entered.

Fault effects

- ♦ Electrical substitute value is assumed
- ♦ Substitute value is assumed
- ♦ All adaptations are deactivated
- ♦ No start/stop

P172F

Diagnosis information PDK control unit

Transmission temperature sensor - short circuit to B+

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit in line between oil temperature sensor and PDK control unit
- ♦ Oil temperature sensor faulty
- ♦ If fault P1775 is additionally entered, then Ground check oil temperature sensor

Fault setting condition(s)

- Measured voltage > threshold value (2.35 V)

Fault effects

- ♦ Substitute value is assumed
- ♦ Clutch pressure < holding threshold 8.1 bar and torque limitation
- ♦ Transmission limp-home display in the instrument cluster control unit

Diagnosis information PDK control unit

Distance sensor information - shift rod 1 outside validity range

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Distance sensor 1 faulty

Fault setting condition(s)

- Distance sensor 1 outside valid value range (10 - 90%)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Hold last gear
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

Fault finding notes

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

Diagnosis information PDK control unit

Distance sensor information - shift rod 2 outside validity range

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Distance sensor 2 faulty

Fault setting condition(s)

- Distance sensor 2 outside valid value range (10 - 90%)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Hold last gear
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

Fault finding notes

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

Diagnosis information PDK control unit

Distance sensor information - shift rod 3 outside validity range

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Distance sensor 3 faulty

Fault setting condition(s)

- Distance sensor 3 outside valid value range (10 - 90%)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Hold last gear
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjuster 4, 5, 6 is switched off
- ♦ Low-side solenoid valve 1, 2 is switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

Fault finding notes

- ♦ Check vehicle wiring harness and internal wiring harness
- ♦ Check distance sensor by moving the shift rods

